DOT AND CROSS PRODUCT

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers [vectors]. The *dot product* [also called the *scalar product*] of z_1 and z_2 is defined as the real number

$$z_1 \circ z_2 = x_1x_2 + y_1y_2 = |z_1|\,|z_2| \cos\theta \tag{1.12}$$

where θ is the angle between z_1 and z_2 which lies between 0 and π.

The *cross product* of z_1 and z_2 is defined as the vector $z_1 \times z_2 = (0, 0, x_1y_2 - y_1x_2)$ perpendicular to the complex plane having magnitude

$$|z_1 \times z_2| = x_1y_2 - y_1x_2 = |z_1||z_2| \sin\theta \tag{1.13}$$

Theorem 1.2 Let z_1 and z_2 be non-zero. Then:

1. A necessary and sufficient condition that z_1 and z_2 be perpendicular is that $z_1 \cdot z_2 = 0$.
2. A necessary and sufficient condition that z_1 and z_2 be parallel is that $|z_1 \times z_2| = 0$.
3. The magnitude of the projection of z_1 on z_2 is $|z_1 \cdot z_2| / |z_2|$.
4. The area of a parallelogram having sides z_1 and z_2 is $|z_1 \times z_2|$.

COMPLEX CONJUGATE COORDINATES

A point in the complex plane can be located by rectangular coordinates (x, y) or polar coordinates (r, θ). Many other possibilities exist. One such possibility uses the fact that $x = \frac{1}{2}(z + \bar{z})$, $y = (1/2i)(z - \bar{z})$ where $z = x + iy$. The coordinates $(z, \bar{z})$ which locate a point are called *complex conjugate coordinates* or briefly *conjugate coordinates* of the point [see Problems 1.43 and 1.44].

POINT SETS

Any collection of points in the complex plane is called a (*two-dimensional*) point set, and each point is called a *member* or *element* of the set. The following fundamental definitions are given here for reference.

1. **Neighbourhoods.** A *delta*, or δ, *neighbourhood* of a point z_0 is the set of all points z such that $|z - z_0| < \delta$ where δ is any given positive number. A *deleted* δ*neighbourhood* of z_0 is a neighbourhood of z_0 in which the point z_0 is omitted, i.e. $0 < |z - z_0| < \delta$.
2. **Limit Points.** A point z_0 is called a *limit point*, *cluster point*, or *point of accumulation* of a point set S if every deleted δ neighbourhood of z_0 contains points of S.

   Since δ can be any positive number, it follows that S must have infinitely many points. Note that z_0 may or may not belong to the set S.
3. **Closed Sets.** A set S is said to be *closed* if every limit point of S belongs to S, i.e. if S contains all its limit points. For example, the set of all points z such that $|z| \leq 1$ is a closed set.
4. **Bounded Sets.** A set S is called *bounded* if we can find a constant M such that $|z| < M$ for every point z in S. An *unbounded set is* one which is not bounded. A set which is both bounded and closed is called *compact*.
5. **Interior, Exterior and Boundary Points.** A point z_0 is called an *interior point* of a set S if we can find a δ neighbourhood of z_0 all of whose points belong to S. If every δ neighbourhood of z_0 contains points belonging to S and also points not belonging to S, then z_0 is called a *boundary point*. If a point is not an interior or boundary point of a set S, it is an *exterior point* of S.
6. **Open Sets.** An *open set* is a set which consists only of interior points. For example, the set of points z such that $|z| < 1$ is an open set.

7. **Connected Sets.** An open set S is said to be *connected* if any two points of the set can be joined by a path consisting of straight line segments (i.e. a *polygonal path*) all points of which are in S.
8. **Open Regions or Domains.** An open connected set is called an *open region* or *domain*.
9. **Closure of a Set.** If to a set S we add all the limit points of S, the new set is called the *closure* of S and is a closed set.
10. **Closed Regions.** The closure of an open region or domain is called a *closed region*.
11. **Regions.** If to an open region or domain we add some, all or none of its limit points, we obtain a set called a *region*. If all the limit points are added, the region is *closed*; if none are added, the region is *open*. In this book whenever we use the word *region* without qualifying it, we shall mean *open region* or *domain*.
12. **Union and Intersection of Sets.** A set consisting of all points belonging to set S_1 or set S_2 or to both sets S_1 and S_2 is called the *union* of S_1 and S_2 and is denoted by $S_1 \cup S_2$.

    A set consisting of all points belonging to both sets S_1 and S_2 is called the *intersection* of S_1 and S_2 and is denoted by $S_1 \cap S_2$.
13. **Complément of a Set.** A set consisting of all points which do not belong to S is called the *complement* of S and is denoted by $\tilde{S}$ or S^c.
14. **Null Sets and Subsets.** It is convenient to consider a set consisting of no points at all. This set is called the *null set* and is denoted by $\varnothing$. If two sets S_1 and S_2 have no points in common (in which case they are called *disjoint* or *mutually exclusive sets*), we can indicate this by writing $S_1 \cap S_2 = \varnothing$.

    Any set formed by choosing some, all or none of the points of a set S is called a *subset* of S. If we exclude the case where all points of S are chosen, the set is called a *proper subset* of S.
15. **Countability of a Set.** If the members or elements of a set can be placed into a one to one correspondence with the natural numbers 1, 2, 3, ..., the set is called *countable* or *denumerable*; otherwise it is *non-countable* or *non-denumerable*.

The following are two important theorems on point sets.

1. **Weierstrass–Bolzano Theorem.** Every bounded infinite set has at least one limit point.
2. **Heine–Borel Theorem.** Let S be a compact set each point of which is contained in one or more of the open sets $A_1, A_2, \ldots$ [which are then said to *cover* S]. Then there exists a finite number of the sets $A_1, A_2, \ldots$ which will cover S.

SOLVED PROBLEMS

FUNCTIONS AND TRANSFORMATIONS

2.1 Let $w = f(z) = z^2$. Find the values of w which correspond to (a) $z = -2 + i$ and (b) $z = 1 - 3i$, and show how the correspondence can be represented graphically.

Solution

(a) $w = f(-2 + i) = (-2 + i)^2 = 4 - 4i + i^2 = 3 - 4i$

(b) $w = f(1 - 3i) = (1 - 3i)^2 = 1 - 6i + 9i^2 = -8 - 6i$

The point $z = -2 + i$, represented by point P in the z plane of Fig. 2.6, has the *image point* $w = 3 - 4i$ represented by P' in the w plane of Fig. 2.7. We say that P is *mapped* into P' by means of the *mapping function* or *transformation* $w = z^2$. Similarly, $z = 1 - 3i$ [point Q of Fig. 2.6] is mapped into $w = -8 - 6i$ [point Q' of Fig. 2.7]. To each point in the z plane there corresponds one and only one point (image) in the w plane, so that w is a single-valued function of z.

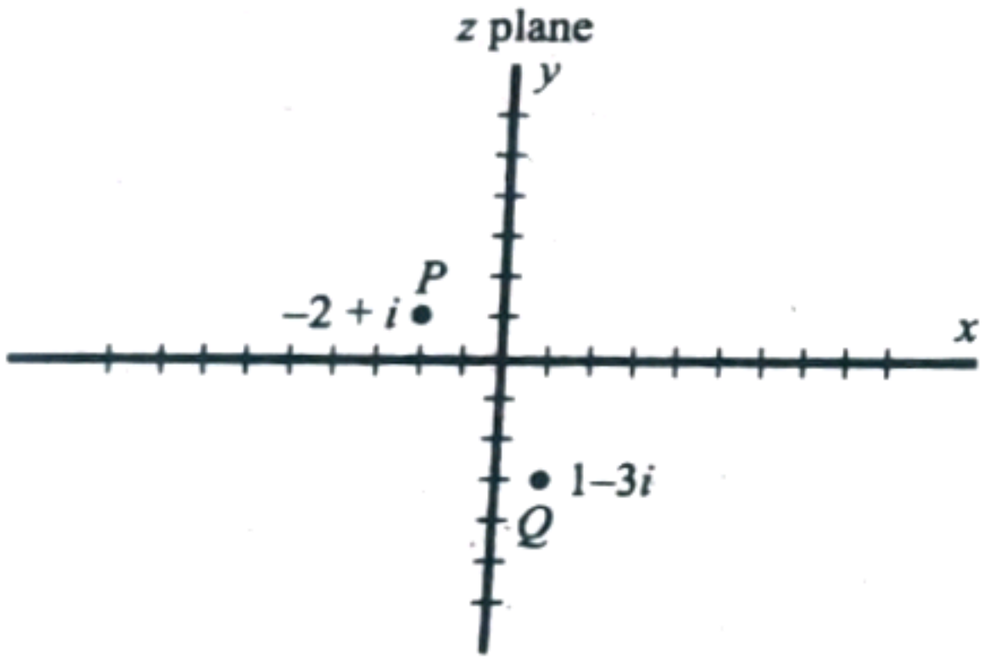


Fig. 2.6

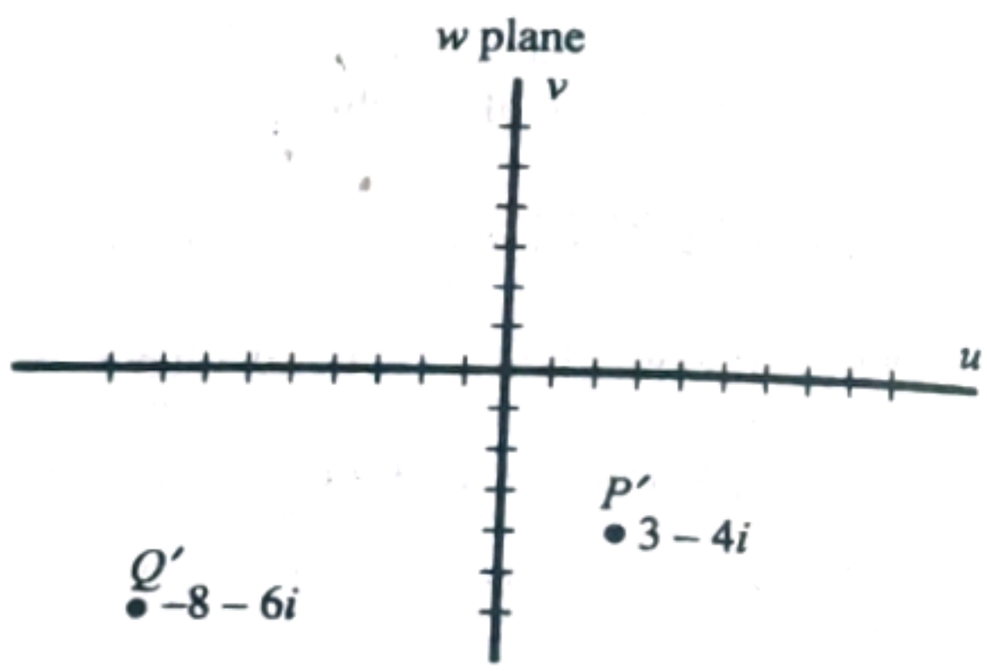


Fig. 2.7

2.2 Show that the line joining the points P and Q in the z plane of Problem 2.1 [Fig. 2.6] is mapped by $w = z^2$ into a curve joining points $P'Q'$ [Fig. 2.7] and determine the equation of this curve.

Solution Points P and Q have coordinates $(-2, 1)$ and $(1, -3)$. Then the parametric equations of the line joining these points are given by

$$\frac{x-(-2)}{1-(-2)} = \frac{y-1}{-3-1} = t \qquad \text{or} \qquad x = 3t - 2,\ y = 1 - 4t$$

The equation of the line PQ can be represented by $z = 3t - 2 + i(1 - 4t)$. The curve in the w plane into which line is mapped has the equation

$$\begin{aligned} w = z^2 &= \{3t - 2 + i(1 - 4t)\}^2 = (3t - 2)^2 - (1 - 4t)^2 + 2(3t - 2)(1 - 4t)i \\ &= 3 - 4t - 7t^2 + (-4 + 22t - 24t^2)i \end{aligned}$$

Then, since $w = u + iv$, the parametric equations of the image curve are given by

$$u = 3 - 4t - 7t^2,\ v = -4 + 22t - 24t^2$$

By assigning various values to the parameter t, this curve may be graphed.

3 A Point P moves in a counterclockwise direction around a circle in the z plane having center at the origin and radius 1. If the mapping function is $w = z^3$, show that when P makes one complete revolution the image P' of P in the w plane makes three complete revolutions in a counterclockwise direction on a circle having centre at the origin and radius 1.

(b) $$\lim_{z\to -2i}\frac{(2z+3)(z-1)}{z^2-2z+4}=\frac{\lim_{z\to -2i}(2z+3)\lim_{z\to -2i}(z-1)}{\lim_{z\to -2i}(z^2-2z+4)}=\frac{(3-4i)(-2i-1)}{4i}=-\frac{1}{2}+\frac{11}{4}i$$

In this case, the limits of the numerator and denominator are each zero and the theorems on limits fail to apply. However, by obtaining the factors of the polynomials, we see that

$$\lim_{z\to 2e^{\pi i/3}}\frac{z^3+8}{z^4+4z^2+16}=\lim_{z\to 2e^{\pi i/3}}\frac{(z+2)(z-2e^{\pi i/3})(z-2e^{5\pi i/3})}{(z-2e^{\pi i/3})(z-2e^{2\pi i/3})(z-2e^{4\pi i/3})(z-2e^{5\pi i/3})}$$

$$=\lim_{z\to 2e^{\pi i/3}}\frac{(z+2)}{(z-2e^{2\pi i/3})(z-2e^{4\pi i/3})}=\frac{e^{\pi i/3}+1}{2(e^{\pi i/3}-e^{2\pi i/3})(e^{\pi i/3}-e^{4\pi i/3})}$$

$$=\frac{3}{8}-\frac{\sqrt{3}}{8}i$$

Another method. Since $z^6-64=(z^2-4)(z^4+4z^2+16)$, the problem is equivalent to finding

$$\lim_{z\to 2e^{\pi i/3}}\frac{(z^2-4)(z^3+8)}{z^6-64}=\lim_{z\to 2e^{\pi i/3}}\frac{z^2-4}{z^3-8}=\frac{e^{2\pi i/3}-1}{2(e^{\pi i}-1)}=\frac{3}{8}-\frac{\sqrt{3}}{8}i$$

2.30 Prove that $\lim_{z\to 0}\frac{\bar{z}}{z}$ does not exist.

Solution If the limit is to exis,t it must be independent of the manner in which z approaches the point 0. Let $z\to 0$ along the x axis. Then $y=0$, and $z=x+iy=x$ and $\bar{z}=x-iy=x$, so that the required limit is

$$\lim_{x\to 0}\frac{x}{x}=1$$

Let $z\to 0$ along the y axis. Then $x=0$, and $z=x+iy=iy$ and $\bar{z}=x-iy=-iy$, so that the required limit is

$$\lim_{x\to 0}\frac{-iy}{iy}=-1$$

Since the two approaches do not give the same answer, the limit does not exist.

CONTINUITY

2.31 (a) Prove that $f(z)=z^2$ is continuous at $z=z_0$.

(b) Prove that $f(z)=\begin{cases}z^2 & z\neq z_0\\ 0 & z=z_0\end{cases}$, where $z_0\neq 0$, is discontinuous at $z=z_0$.

Solution

(a) By Problem 2.23(a), $\lim_{z\to z_0}f(z)=f(z_0)=z_0^2$ and so $f(z)$ is continuous at $z=z_0$.

Another method. We must show that given any $\epsilon>0$, we can find $\delta>0$ (depending on ϵ) such that $|f(z)-f(z_0)|=|z^2-z_0^2|<\epsilon$ when $|z-z_0|<\delta$. The proof patterns that given in Problem 2.23(a).

(b) By Problem 2.23(b), $\lim_{z\to z_0}f(z)=z_0^2$, but $f(z_0)=0$. Hence $\lim_{z\to z_0}f(z)\neq f(z_0)$ and so $f(z)$ is discontinuous at $z=z_0$ if $z_0\neq 0$.

If $z_0=0$, then $f(z)=0$; and since $\lim_{z\to z_0}f(z)=0=f(0)$, we see that the function is continuous.

Solution

If $f(z)$ is analytic in $\mathcal{R}$ then the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \tag{1}$$

and

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \tag{2}$$

are satisfied in $\mathcal{R}$. Assuming u and v have continuous second partial derivatives, we can differentiate both sides of (1) with respect to x and (2) with respect to y to obtain

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x\,\partial y} \tag{3}$$

and

$$\frac{\partial^2 v}{\partial y\,\partial x} = -\frac{\partial^2 u}{\partial y^2} \tag{4}$$

from which

$$\frac{\partial^2 u}{\partial x^2} = -\frac{\partial^2 u}{\partial y^2} \text{ or } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0\,,$$

i.e. u is harmonic.

Similarly, by differentiating both sides of (1) with respect to y and (2) with respect to x, we find

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

and v is harmonic.

It will be shown later (Chapter 5) that if $f(z)$ is analytic in $\mathcal{R}$, all its derivatives exist and are continuous in $\mathcal{R}$. Hence, the above assumptions will not be necessary.

3.7 (a) Prove that $u = e^{-x}(x \sin y - y \cos y)$ is harmonic.

(b) Find v such that $f(z) = u + iv$ is analytic.

Solution

(a) $\dfrac{\partial u}{\partial x} = (e^{-x})(\sin y) + (-e^{-x})(x \sin y - y \cos y) = e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}(e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y) = -2e^{-x} \sin y + xe^{-x} \sin y - ye^{-x} \cos y \tag{1}$$

$$\frac{\partial u}{\partial y} = e^{-x}(x \cos y + y \sin y - \cos y) = xe^{-x} \cos y + ye^{-x} \sin y - e^{-x} \cos y$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}(xe^{-x} \cos y + ye^{-x} \sin y - e^{-x} \cos y) = -xe^{-x} \sin y + 2e^{-x} \sin y + ye^{-x} \cos y \tag{2}$$

Adding (1) and (2) yields $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$ and u is harmonic.

(b) From the Cauchy–Riemann equations,

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y \tag{3}$$

Integrating with respect to r, we obtain $u = -r^2 \sin 2\theta + r \sin \theta + \phi(\theta)$, where $\phi(\theta)$ is an arbitrary function.

$$\therefore \quad \frac{\partial u}{\partial \theta} = -2r^2 \cos 2\theta + r \cos \theta + \phi'(\theta) \tag{3}$$

From (2) and (3), we get

$$-2r^2 \cos 2\theta + r \cos \theta + \phi'(\theta) = -2r^2 \cos 2\theta + r \cos \theta$$

$$\therefore \quad \phi'(\theta) = 0$$

or $$\phi(\theta) = C$$

Thus, $u = -r^2 \sin 2\theta + r \sin \theta + C$

Hence, $f(z) = u + iv$

$$= (-r^2 \sin 2\theta + r \sin \theta + C) + i(r^2 \cos 2\theta - r \cos \theta + 2)$$
$$= r^2 (-\sin 2\theta + i \cos 2\theta) + r(\sin \theta - i \cos \theta) + C + 2i$$
$$= ir^2 (\cos 2\theta + i \sin 2\theta) - ir(\cos \theta - i \sin \theta) + C + 2i$$
$$= ir^2 e^{i2\theta} - ire^{-i\theta} + C + 2i$$
$$= i(r^2 e^{i2\theta} - re^{-i\theta}) + C + 2i$$

3.16 Find the values of constants a, b, c and d such that the function $f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$ is analytic.

Solution

$$f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2) = u + iv(\text{say})$$

where

$$u = x^2 + axy + by^2, \quad v = cx^2 + dxy + y^2$$

$\therefore$

$$u_x = 2x + ay, \quad u_y = ax + 2by$$
$$v_x = 2cx + dy, \quad v_y = dx + 2y$$

Since $f(z) = u + iv$ is analytic, so Cauchy-Riemann equation must be satisfied.

i.e., $u_x = v_y$ and $u_y = -v_x$

Now,

$$u_x = v_y \Rightarrow 2x + ay = dx + 2y \tag{1}$$

and

$$u_y = -v_x \Rightarrow ax + 2by = -2cx - dy \tag{2}$$

$$(1) \Rightarrow 2x - dx + ay - 2y = 0 \Rightarrow (2 - d)x + (a - 2)y = 0$$
$$(2) \Rightarrow ax + 2cx + 2by + dy = 0 \Rightarrow (a + 2c)x + (2b + d)y = 0$$

(1) and (2) will hold good if

and

$$2 - d = 0, a - 2 = 0$$
$$a + 2c = 0, 2b + d = 0$$

i.e.,

$$a = 2, d = 2, c = -1, b = -1$$

3.17 Show that the function $f(z) = \sin x \cosh y + i \cos x \sinh y$ is continuous as well as analytic everywhere.

Solution If

$$f(z) = u(x, y) + iv(x, y)$$

then

$$u(x, y) = \sin x \cosh y, \quad \text{and} \quad v(x, y) = \cos x \sinh y$$

Since u and v both are rational functions of x and y, whose denominators are non-zero for all values of x and y, therefore u and v are both continuous everywhere. Hence $f(z) = u + iv$ is also continuous everywhere.

Again,

$$\frac{\partial u}{\partial x} = \cos x \cosh y, \quad \frac{\partial u}{\partial y} = \sin x \sinh y$$

$$\frac{\partial v}{\partial x} = -\sin x \sinh y, \quad \frac{\partial v}{\partial y} = \cos x \cosh y.$$

The four partial derivatives are rational functions of x and y with non-zero denominators for all values of x and y, therefore, they are continuous everywhere.

Also, here, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

That is, Cauchy-Riemann conditions are satisfied.

Thus the four partial derivatives being continuous everywhere and Cauchy-Riemann equations being satisfied. The function $f(z)$ is analytic everywhere.

.18 Prove that the function $f(z) = u + iv$, where

$$f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2}(z \neq 0), \quad f(0) = 0.$$

Is continuous and that Cauchy-Riemann equations are satisfied at the origin, yet $f'(z)$ does not exist there

Solution Here, $$f(z) = u + iv = \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2}$$

So $$u = \frac{x^3 - y^3}{x^2 + y^2}, \quad v = \frac{x^3 + y^3}{x^2 + y^2} \text{ (where } z \neq 0).$$

Here, we see that both u and v are rational and finite for all values of $z \neq 0$, so u and v are continuous at all those points for which $z \neq 0$. Hence $f(z)$ is continuous where $z \neq 0$.

At the origin $u = 0$, $v = 0$. [since $f(0) = 0$].

Hence u and v are both continuous at the origin; therefore $f(z)$ is continuous at the origin.

Now, at the origin

$$\frac{\partial u}{\partial x} = \lim_{x \to 0} \frac{u(x,0) - u(0,0)}{x} = \lim_{x \to 0}\left(\frac{x}{x}\right) = 1, \quad \frac{\partial u}{\partial y} = \lim_{y \to 0} \frac{u(0,y) - u(0,0)}{y} = \lim_{y \to 0}\left(\frac{-y}{+y}\right) = -1$$

$$\frac{\partial v}{\partial x} = \lim_{x \to 0} \frac{v(x,0) - v(0,0)}{x} = \lim_{x \to 0}\left(\frac{x}{x}\right) = 1, \quad \frac{\partial v}{\partial y} = \lim_{y \to 0} \frac{v(0,y) - v(0,0)}{y} = \lim_{y \to 0}\left(\frac{y}{y}\right) = +1$$

$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. Therefore, Cauchy-Riemann equations are satisfied at $z = 0$. Again,

$$f'(0) = \lim_{z \to 0} \frac{f(z) - f(0)}{z} = \lim_{z \to 0}\left[\frac{x^3 - y^3 + i(x^3 + y^3)}{x^2 + y^2} \cdot \frac{1}{x + iy}\right]$$

Let $z \to 0$ along $y = x$ then we have

$$f'(0) = \lim_{z \to 0} \frac{x^3 - y^3 + i(x^3 + y^3)}{x^2 + y^2} \cdot \frac{1}{x + iy} = \lim_{z \to 0} \frac{2i}{2(1+i)} = \frac{1}{2}(1 - i).$$

urther, let $z \to 0$ along $y = 0$, then we have $f'(0) = \lim_{z \to 0} \frac{x^3(1+i)}{x^3} = 1 + i$. Hence, $f'(0)$ is not unique. hus, $f'(z)$ does not exist at the origin.

5.6 Evaluate $\int_C \frac{e^{-z}}{z+1}\,dz$, where C is the circle $|z| = \frac{1}{2}$.

Solution Here, $f(z) = e^{-z}$ is an analytic function.

The point $z = -1$ lies outside the circle $|z| = \frac{1}{2}$.

$\therefore$ The function $\frac{e^{-z}}{z+1}$ is analytic within and on C.

By Cauchy's theorem, we have $\int_C \frac{e^{-z}}{z+1}\,dz = 0$.

5.7 Evaluate $\int_C \frac{3z^2+z}{z^2-1}\,dz$, where C is the circle $|z-1| = 1$.

Solution The integrand has singularities, where $z^2 - 1 = 0$ i.e. at $z = 1$ and $z = -1$.
The circle $|z-1| = 1$ has center at $z = 1, f(z) = 3z^2 + z$, is an analytic function.

Also,
$$\frac{1}{z^2-1} = \frac{1}{(z-1)(z+1)} = \frac{1}{2}\left(\frac{1}{z-1} - \frac{1}{z+1}\right)$$

$\therefore$
$$\int_C \frac{3z^2+z}{z^2-1}dz = \frac{1}{2}\int_C \frac{3z^2+z}{z-1}\,dz - \frac{1}{2}\int_C \frac{3z^2+z}{z+1}\,dz$$

By Cauchy's integral formula,

$$\int_C \frac{3z^2+z}{z-1}\,dz = 2\pi i f(1) = 8\pi i \qquad \text{where } f(z) = 3z^2$$

By Cauchy's theorem, $\int_C \frac{3z^2+z}{z+1}\,dz = 0$

$\therefore$ From (1), we have $\int_C \frac{3z^2+z}{z^2-1}\,dz = 4\pi i$.

5.8 Prove Cauchy's integral formula for multiply-connected regions.

Solution

We present a proof for the multiply-connected region $\mathcal{R}$ bounded by the simple closed curves C_1 and C_2 as indicated in Fig. 5.4. Extensions to other multiply-connected regions are easily made (see Problem 5.49).

Construct a circle Γ having centre at any point a in $\mathcal{R}$ so that Γ lies entirely in $\mathcal{R}$. Let $\mathcal{R}'$ consist of the set of points in $\mathcal{R}$ which are exterior to Γ. Then, the function $\frac{f(z)}{z-a}$ is analytic inside and on the boundary of $\mathcal{R}'$. Hence, by Cauchy's theorem for multiply-connected regions (Problem 4.15, Chapter 4),

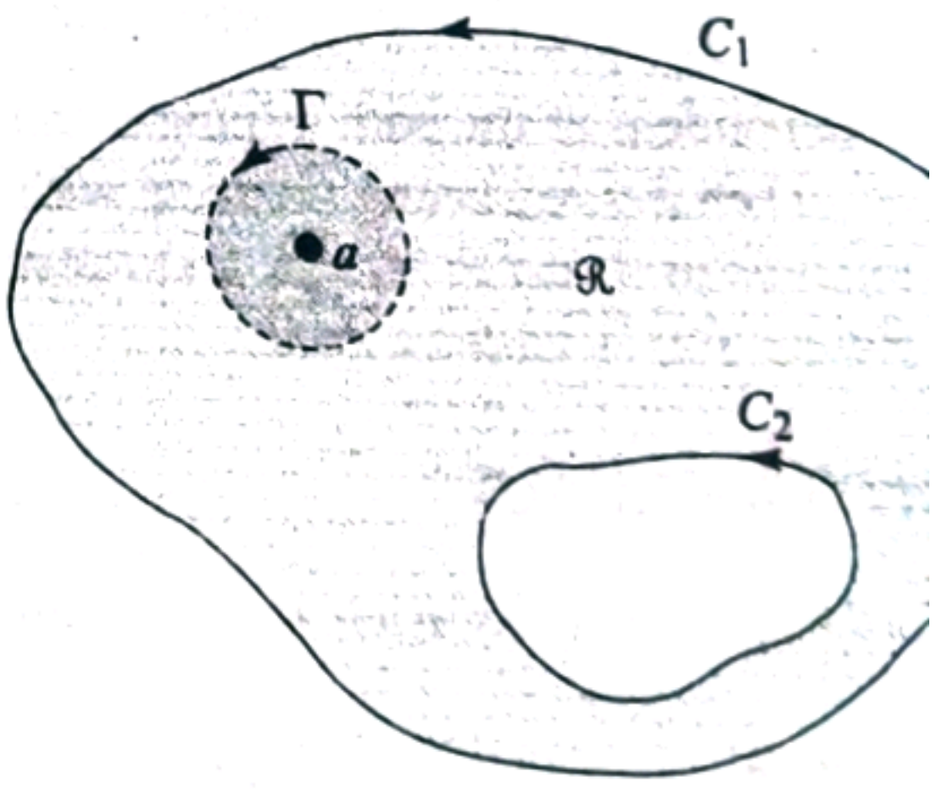


Fig. 5.4

6.2 (a) Prove that the series $z(1-z)+z^2(1-z)+z^3(1-z)+\ldots$ converges for $|z|<1$, and (b) find its sum.

Solution The sum of the first n terms of the series is

$$\begin{aligned} S_n(z) &= z(1-z)+z^2(1-z)+\ldots+z^n(1-z) \\ &= z-z^2+z^2-z^3+\ldots+z^n-z^{n+1} \\ &= z-z^{n+1} \end{aligned}$$

Now $|S_n(z)-z|=|-z^{n+1}|=|z|^{n+1}<\epsilon$ for $(n+1)\ln|z|<\ln\epsilon$, i.e. $n+1>\dfrac{\ln\epsilon}{\ln|z|}$ or $n>\dfrac{\ln\epsilon}{\ln|z|}-1$ if $z\neq 0$.

If $z=0$, $S_n(0)=0$ and $|S_n(0)-0|<\varepsilon$ for all n.

Hence $\lim_{n\to\infty} S_n(z)=z$, the required sum for all z such that $|z|<1$.

Another method.

Since $S_n(z)=z-z^{n+1}$, we have [by Problem 2.41, Chapter 2. in which we showed that $\lim_{n\to\infty} z^n=0$ if $|z|<1$]

$$\text{Required sum} = S(z)=\lim_{n\to\infty} S_n(z)=\lim_{n\to\infty}(z-z^{n+1})=z$$

ABSOLUTE AND UNIFORM CONVERGENCE

6.3 (a) Prove that the series in Problem 6.2 converges uniformly to the sum z for $|z|\leq\frac{1}{2}$.

(b) Does the series converge uniformly for $|z|\leq 1$? Explain.

Solution

(a) In Problem 6.2, we have shown that $|S_n(z)-z|<\epsilon$ for all $n>\dfrac{\ln\epsilon}{\ln|z|}-1$, i.e. the series converges to the sum z for $|z|<1$ and thus for $|z|\leq\frac{1}{2}$.

Now if $|z|\leq\frac{1}{2}$, the largest value of $\dfrac{\ln\epsilon}{\ln|z|}-1$ occurs where $|z|=\frac{1}{2}$ and is given by $\dfrac{\ln\epsilon}{\ln(1/2)}-1=N$. It follows that $|S_n(z)-z|<\epsilon$ for all $n>N$ where N depends only on ε and not on the particular z in $|z|\leq\frac{1}{2}$. Thus the series converges uniformly to z for $|z|\leqq\frac{1}{2}$.

(b) The same argument given in part (a) serves to show that the series converges uniformly to sum z for $|z|\leq .9$ or $|z|\leq .99$ by using $N=\dfrac{\ln\epsilon}{\ln(.9)}-1$ and $N=\dfrac{\ln\epsilon}{\ln(.99)}-1$ respectively.

However, it is clear that we cannot extend the argument to $|z|\leq 1$ since this would require $N=\dfrac{\ln\epsilon}{\ln 1}-1$ which is infinite, i.e., there is no finite value of N which can be used in this case. Thus, the series does not converge uniformly for $|z|\leq 1$.

6.4 (a) Prove that the sequence $\left\{\dfrac{1}{1+nz}\right\}$ is uniformly convergent to zero for all z such that $|z|\geq 2$.

(b) Can the region of uniform convergence in (a) be extended? Explain.

Solution

(a) We have $\left|\dfrac{1}{1+nz}-0\right|<\epsilon$ when $\dfrac{1}{|1+nz|}<\epsilon$ or $|1+nz|>1/\epsilon$. Now $|1+nz|\leq|1|+|nz|=1+$

$$a_n = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^{n+1}}\,dz \qquad n = 0, \pm 1, \pm 2, \ldots \tag{6.7}$$

With an appropriate change of notation, we can write the above as

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \ldots + \frac{a_{-1}}{z-a} + \frac{a_{-2}}{(z-a)^2} + \ldots \tag{6.8}$$

where

$$a_n = \frac{1}{2\pi i}\oint_C \frac{f(\zeta)}{(\zeta-a)^{n+1}}\,d\zeta \quad n = 0, \pm 1, \pm 2, \ldots \tag{6.9}$$

This is called *Laurent's theorem* and (6.5) or (6.8) with coefficients (6.6), (6.7) or (6.9) is called a *Laurent series* or *expansion*.

The part $a_0 + a_1(z-a) + a_2(z-a)^2 + \ldots$ is called the *analytic part* of the Laurent series, while the remainder of the series, which consists of inverse powers of $z-a$ is called the *principal part*. If the principal part is zero, the Laurent series reduces to a Taylor series.

CLASSIFICATION OF SINGULARITIES

It is possible to classify the singularities of a function $f(z)$ by examination of its Laurent series. For this purpose, we assume that in Fig. 6.1, $R_2 = 0$, so that $f(z)$ is analytic inside and on C_1 except at $z = a$, which is an isolated singularity [see Page 3.5]. In the following, all singularities are assumed isolated unless otherwise indicated.

1. **Poles.** If $f(z)$ has the form (6.8) in which the principal part has only a finite number of terms given by

$$\frac{a_{-1}}{z-a} + \frac{a_{-2}}{(z-a)^2} + \cdots + \frac{a_{-n}}{(z-a)^n}$$

where $a_{-n} \neq 0$, then $z = a$ is called a *pole of order n*. If $n = 1$, it is called a *simple pole*.

If $f(z)$ has a pole at $z = a$, then $\lim_{z\to a} f(z) = \infty$ [see Problem 6.32].

2. **Removable singularities.** If a single-valued function $f(z)$ is not defined at $z = a$ but $\lim_{z\to a} f(z)$ exists, then $z = a$ is called a *removable singularity*. In such case we define $f(z)$ at $z = a$ as equal to $\lim_{z\to a} f(z)$, and $f(z)$ will then be analytic at a.

Example 6.1 If $f(z) = \sin z/z$, then $z = 0$ is a removable singularity since $f(0)$ is not defined but $\lim_{z\to a} \sin z/z = 1$. We define $f(0) = \lim_{z\to a} \sin z/z = 1$. Note that in this case

$$\frac{\sin z}{z} = \frac{1}{z}\left\{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots\right\} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \frac{z^6}{7!} + \cdots$$

3. **Essential singularities.** If $f(z)$ is single-valued, then any singularity which is not a pole or removabl singularity is called an *essential singularity*. If $z = a$ is an essential singularity of $f(z)$, the principal pa of the Laurent expansion has infinitely many terms.

Example 6.2 Since $e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \cdots$, $z = 0$ is an essential singularity.

The following two related theorems are of interest (see Problems 6.154–6.156):

Casorati–Weierstrass theorem. In any neighbourhood of an isolated essential singularity a, an otherwise analytic function $f(z)$ comes arbitrarily close to any complex number. In symbols, given any positive numbers δ and ϵ and any complex number A, there exists a value of z inside the circle $|z-a| = \delta$ for which $|f(z) - A| < \epsilon$.

Picard's theorem. In the neighbourhood of an isolated essential singularity a, an otherwise analytic function $f(z)$ can take on any value whatsoever with perhaps one exception.

4. **Branch points.** A point $z = z_0$ is called a *branch point* of the multiple-valued function $f(z)$ if the branches of $f(z)$ are interchanged when z describes a closed path about z_0 [see Page 2.5]. A branch point is a non-isolated singularity. Since each of the branches of a multiple-valued function is analytic, all of the theorems for analytic functions, in particular Taylor's theorem, apply.

   Example 6.3 The branch of $f(z) = z^{1/2}$, which has the value 1 for $z = 1$, has a Taylor series of the form $a_0 + a_1(z-1) + a_2(z-1)^2 + \ldots$ with radius of convergence $R = 1$ [the distance from $z = 1$ to the nearest singularity, namely the branch point $z = 0$].

5. **Singularities at infinity.** By letting $z = 1/w$ in $f(z)$, we obtain the function $f(1/w) = F(w)$. Then the nature of the singularity for $f(z)$ at $z = \infty$ [the point at infinity] is defined to be the same as that of $F(w)$ at $w = 0$.

   Example 6.4 $f(z) = z^3$ has a pole of order 3 at $z = \infty$, since $F(w) = f(1/w) = 1/w^3$ has a pole of order 3 at $w = 0$. Similarly $f(z) = e^z$ has an essential singularity at $z = \infty$, since $F(w) = f(1/w) = e^{1/w}$ has an essential singularity at $w = 0$.

ENTIRE FUNCTIONS

A function which is analytic everywhere in the finite plane [i.e. everywhere except at ∞] is called an *entire function* or *integral function*. The functions e^z, $\sin z$, $\cos z$ are entire functions.

An entire function can be represented by a Taylor series which has an infinite radius of convergence. Conversely if a power series has an infinite radius of convergence, it represents an entire function.

Note that by Liouville's theorem [Chapter 5, Page 5.2] a function which is analytic *everywhere including* ∞ must be a constant.

MEROMORPHIC FUNCTIONS

A function that is analytic everywhere in the finite plane except at a finite number of poles is called a *meromorphic function*.

Example 6.5 $\dfrac{z}{(z-1)(z+3)^2}$ which is analytic everywhere in the finite plane except at the poles $z = 1$ (simple pole) and $z = -3$ (pole of order two) is a meromorphic function.

LAGRANGE'S EXPANSION

Let z be that root of $z = a + \zeta\phi(z)$ which has the value $z = a$ when $\zeta = 0$ Then, if $\phi(z)$ is analytic inside and on a circle C containing $z = a$, we have

$$z = a + \sum_{n=1}^{\infty} \frac{\zeta^n}{n!} \frac{d^{n-1}}{da^{n-1}} \{[\phi(a)]^n\} \tag{6.10}$$